

Class-D Homework Assignment 3A

Power Stage Design

- In Cadence Virtuoso create with elements from **analogLib**, **ahdlLib** technology library **tsmc018** a class-D output stage with NMOS powerFETs and inverter gate drivers as shown on slide 6 of the lecture.
 - use one voltage source to create a single power supply of +12V
 - use 5V voltage sources as gate driver supplies
 - for the powerFETs use device **nld12_g5a_nbl_mac**. Determine with simulations the device size (gate width / number of fingers) needed to achieve an on-resistance of 50m Ω at 150°C.
 - determine with a transient simulation the gate charge Q_G and threshold voltage V_{TH} of the powerFETs.
 - create gate drivers (inverter) with 5V devices **nmos5v_mac** and **pmos5v_mac**. Determine the size of the devices such that the rise and fall time is about **25ns**. Make sure that shoot-through is avoided.
 - use voltage controlled voltage sources (**vcvs**) as levelshifters to drive the gate drivers
 - use **5ns dead time** to drive lowside and highside. Tip: use the logic gates from **ahdlLib** (with suitable parameters) to derive the proper signals for the gate drivers from a PWM signal.
 - verify that the behavior is correct with load currents of +/- 2A
- Connect the LC-filter from Homework Assignment 1A.
 - make a copy of the power stage and LC-filter to create a bridge-tied-load BTL configuration.
 - connect a 4 Ω load.
 - create two PWM generators that produces AD-PWM signals with $F_{PWM}=1\text{MHz}$
 - make a plot of the differential-mode and common-mode output voltage after the LC-filter at zero modulation index: Describe and explain your observations
 - make a plot of efficiency versus modulation index. (Caveat: don't forget the current in the gate drivers)
- Increase the gate driver size by a factor of four and repeat the plot of efficiency versus modulation index
 - describe and explain your observations

Class-D Homework Assignment 3B

Power Stage Design

- In Cadence Virtuoso create with elements from **analogLib**, **ahdlLib** technology library **tsmc018** a class-D output stage with NMOS powerFETs and inverter gate drivers as shown on slide 6 of the lecture.
 - use two voltage sources to create a symmetrical power supply of +/- 12V
 - use 5V voltage sources as gate driver supplies
 - for the powerFETs use device **nld24_g5b_mac**. Determine with simulations the device size (gate width / number of fingers) needed to achieve an on-resistance of 100m Ω at 150°C.
 - determine with a transient simulation the gate charge Q_G and threshold voltage V_{TH} of the powerFETs.
 - create gate drivers (inverter) with 5V devices **nmos5v_mac** and **pmos5v_mac**. Determine the size of the devices such that the rise and fall time is about 10ns. Make sure that shoot-through is avoided.
 - use voltage controlled voltage sources (**vcvs**) as levelshifters to drive the gate drivers
 - create a PWM generators that produces a PWM signal with $F_{PWM}=1\text{MHz}$
 - use **10ns dead time** to drive lowside and highside. Tip: use the logic gates from **ahdlLib** (with suitable parameters) to derive the proper signals for the gate drivers from a PWM signal.
 - verify that the behavior is correct with load currents of +/- 3A
 - make a plot of the average output voltage V_{PWM} versus modulation index
- Connect the LC-filter and 4 Ω load from Homework Assignment 1A
 - repeat the plot of the average output voltage V_{PWM} versus modulation index. Describe and explain your observations
 - increase the dead time to 50ns and repeat. Describe and explain your observations

Class-D Homework Assignment 3C

Power Stage Design

- In Cadence Virtuoso create with elements from **analogLib**, **ahdlLib** technology library **tsmc018** a class-D output stage with complementary powerFETs and inverter gate drivers as shown on slide 89 of the lecture.
 - use one voltage sources to create a single power supply of +5V
 - for the powerFETs and gate drivers use devices **nmos5v_mac** and **pmos5v_mac**. determine with simulations the device sizes (gate width / number of fingers) needed to achieve an on-resistance of 75m Ω at 150°C.
 - determine with a transient simulation the gate charges Q_G and threshold voltage V_{TH} of the powerFETs.
 - create gate drivers (inverter) with 5V devices **nmos5v_mac** and **pmos5v_mac**. determine the size of the devices such that the rise and fall time is about 5ns. Make sure that shoot-through is avoided.
 - use **10ns dead time** to drive lowside and highside. Tip: use the logic gates from **ahdlLib** (with suitable parameters) to derive the proper signals for the gate drivers from a PWM signal.
 - verify that the behavior is correct with load currents of +/-1.5A
 - resize the gate drivers such that the minimal (soft switching) rise and fall time is 2ns.
- Make a copy of the power stage to create a bridge-tied-load BTL configuration.
 - connect a 4 Ω load consisting of a series connection of a 4 Ω resistor and a 40 μ H inductor.(without LC-filter)
 - create two PWM generators that produces AD-PWM signals with $F_{PWM}=256\text{kHz}$
 - determine the average supply current
 - modify the PWM generators to produce BD-PWM
 - determine the average supply current. describe and explain your observations.